

Invoice PDF: what is left to do

E-commerce v2 | Updated 20 September 2026 | After rounds A, B and C

1. Where things stand

The base feature works: an invoice is created when an order is placed (storefront, admin and walk-in orders), numbered like SI26/1, saved as a frozen copy and drawn as a PDF that follows your Tally sample. Customers download their own invoice, admins download any, controlled by the PDF switch in Website Master and Company Master plus the Invoice tab in Company Settings.

Added in this round:

- Download buttons now appear only when a download will actually work (feature on, order invoiceable). Before, the button showed and then said "not enabled".
- Original and Duplicate copies: a Company Settings switch prints both labelled copies in the same PDF.
- Round-off: a Company Settings switch rounds the invoice total to a whole amount and shows a "Round off" row (off by default).
- Email: a Company Settings switch emails the customer their invoice PDF right after they order (off by default).
- Cancellations: when an order is cancelled by the customer or rejected by an admin, its invoice is marked VOID (stamped CANCELLED) and a full credit note (for example CN26/1) is issued. Both can be downloaded. A cancelled order that never had an invoice no longer gets one.
- Automated tests: the project now has a test command (npm test in the backend, 19 tests) covering amount in words, totals, rounding, tax grouping and PDF rendering.
- A bug outside invoices was fixed: rejecting an already-rejected order used to put the stock back a second time. It is now refused.

Correction to an earlier statement: I said the email service had no attachment support. That was wrong. Nodemailer and SendGrid already support attachments; only SES does not. Emailing therefore needed far less work than I estimated.

2. Do this before using it (setup, no coding)

#	Task	Where	Why it matters
1	Fill in company name, address, phone and email	Company Settings, General tab	Until then the invoice header falls back to the admin name and prints no address or phone.
2	Enter your TRN, invoice prefix and declaration text	Company Settings, Invoice tab	Without a TRN the document is titled "Invoice" instead of "Tax Invoice".
3	Enter bank details if they should appear on invoices	Company Settings, Payment & Bank tab	The bank block only prints when bank details exist (and the vendor is entitled to bank details).
4	To email invoices: assign an email service and add its credentials	Company Master / Website Master, plus backend .env	Your vendor has no email service assigned and no SendGrid or SMTP credentials, so invoice emails would be skipped (logged, order unaffected) until this is done. The SES provider cannot send attachments; use Nodemailer or SendGrid.
5	Then switch on "Email the invoice when an order is placed"	Company Settings, Invoice tab	Off by default. Your account already allows attachments, PDF files (up to 5 MB) and CC/ BCC. Note your settings hold CC/BCC addresses, which the email service only accepts while the CC/BCC feature stays on.
6	Turn on isPDFDownloadableFeatureOn in	Database / seed data	It is on for your current vendor. Vendors without it get a "feature not enabled" message.

#	Task	Where	Why it matters
	Company Master for every other vendor		
7	Move Sales_SI26_1922.pdf out of frontend/public	frontend/public	Files in public/ are served to anyone. That sample contains real bank details.
8	Run node scripts/ migrateOrderCartIdIndex.js on each environment	backend	This database still has the old order index, so only ONE admin or walk-in order can exist at a time.
9	Remember the master-data cache when Redis is on	Deployment	Website/Company Master flag changes can take up to an hour to apply (not an issue locally, Redis is off).

3. Not built yet

Size is a rough guide to effort: Small is under a day, Medium is a few days, Large is a week or more.

#	Feature	Why it matters	What it needs	Size
1	Credit notes for partial returns and exchanges	Only a full cancellation raises a credit note today. A customer who returns part of an order still keeps the whole invoice.	Hook into the return and exchange flows; a credit note that lists only the returned items and their tax; numbering shared with full credit notes.	Large
2	Units, per-line discount % and item codes (HSN/SAC)	The sample shows quantities such as "5 PKTS", a rate per unit, a Disc % column, and tax authorities may expect item codes.	Store the unit and code on each order item when the order is placed; spread the order-level discount across lines; new table columns.	Medium
3	More invoice fields	Payment method and status ("Paid", "Cash on delivery"), payment terms, date of supply, buyer order number and dispatch details appear on the sample.	Extra snapshot fields and inputs. Payment status changes after the order is placed, so this needs a small design decision.	Medium
4	Admin Invoices screen	No place to search, re-download or bulk-download invoices. (The invoice number now shows on the admin order screen, but not in the order list.)	A new admin module (list, search, download) with a module entry.	Medium
5	Email extras	The invoice email is sent once, in the background, with fixed wording. There is no resend button, no retry if the mail server is down, and no email when a credit note is issued.	A "send again" action, a retry, an email for credit notes, and optionally the vendor's own email template instead of the built-in text.	Medium
6	Look and wording per vendor	Colours, a footer note, a terms block, or more than one template.	Template options in Company Settings and a more flexible renderer.	Medium
7	Database-backed automated tests	The 19 automated tests cover logic and PDF drawing. Numbering, who may download what, voiding and emailing were verified by scripts against a real database that were not saved.	A test database setup, then tests for those flows.	Medium

#	Feature	Why it matters	What it needs	Size
8	Invoices for older orders	All 16 current orders were placed before line items were stored. They have no tax breakdown, so they cannot be invoiced correctly.	A decision: leave them uninvoiced (recommended) or create them by hand. Rebuilding from old carts would show wrong tax.	Decision
9	Arabic / bilingual invoice	Many UAE businesses issue bilingual invoices. The current renderer supports Latin text only; Arabic characters would print blank.	A different renderer that can shape right-to-left text, for example an HTML template rendered by headless Chromium (about 170 MB), plus a bilingual layout.	Large

4. Things to know about how invoices work today

- Tax is worked out on the full line amount, before any discount (this is the cart logic, not the invoice). A discount therefore lowers the total but not the VAT. The invoice shows the order exactly as it was charged. Please confirm the rule with your accountant; fixing it means changing the cart, not the invoice.
- Round-off changes only the invoice. Online card payments still charge the exact order amount, so turn round-off on only if you collect payment against the invoice total (cash or bank transfer).
- Tax-inclusive prices are not supported. The tax setup has an inclusive/exclusive option, but the cart always adds tax on top. None of your current taxes use the inclusive option.
- Flat-amount taxes (a fixed sum instead of a percentage) exist in the tax setup but are not handled on the invoice: the amount would print in the percentage column. None are configured today; your GST 18% and VAT 5% are percentage taxes.
- Invoice numbers restart every calendar year (not fiscal year), and so do credit note numbers. Changing the prefix only affects invoices issued afterwards. Credit notes always use the prefix CN.
- The invoice date is the day it was issued. An invoice created late (first download of an order that missed issuing) carries the day of that download.
- A voided invoice keeps its number and stays downloadable, stamped CANCELLED with a reference to its credit note. Whether the tax authority also wants the original invoice kept as issued is worth confirming with your accountant.
- The logo is downloaded from your image storage each time a PDF is made (4 second limit, PNG or JPEG only, cached in memory). If it cannot be fetched, the invoice prints without it.
- In the rare case two people trigger a late issue for the same order at the same instant, one invoice number can be skipped. Normally invoices are issued at placement, so this is very unlikely.
- One check is still pending: the storefront (customer) order page buttons were not clicked in a browser, because that needs a customer login. They share their code with the admin buttons, which were tested in the browser, and the customer downloads were tested against the real server.

5. Suggested order of work

Round	Theme	Contents
Round D	Richer invoice	Units, per-line discount and item codes; extra fields (payment, terms, dispatch).
Round E	Management	Admin Invoices screen; email extras; per-vendor look; database-backed tests.
Round F	Returns	Credit notes for partial returns and exchanges.
Later / optional	Only if needed	Arabic / bilingual invoice; anything for older orders.

6. Decisions needed from you

1. Partial returns: should every return or exchange raise its own credit note automatically?
2. Email: should the customer also get an email when a credit note is issued?
3. Arabic: do your invoices need to be bilingual? (This is the largest remaining item.)
4. Older orders: agree to leave the 16 existing orders without invoices?
5. Tax and discounts: is it correct that VAT is charged on the amount before discount? Please check with your accountant.
6. Voided invoices: does your accountant want the original invoice kept downloadable exactly as issued, as it is now?